

මාපාංක ශ්‍රිත (Modulus Function)

මාපාංක ශ්‍රිත (Modulus Functions)

◆ තාත්වික සංඛ්‍යාවක ලකුණ කුමක් වුවත්, එහි සෘණ නොවන විශාලත්වය මාපාංක ශ්‍රිතය මගින් ලබා දෙයි.

The modulus function outputs the non – negative magnitude of a real number regardless of its sign.

◆ තාත්වික සංඛ්‍යාවක මාපාංකය (හෝ නිරපේක්ෂ අගය) සංඛ්‍යා රේඛාවේ ශුන්‍යයේ සිට එහි දුර නියෝජනය කරයි.

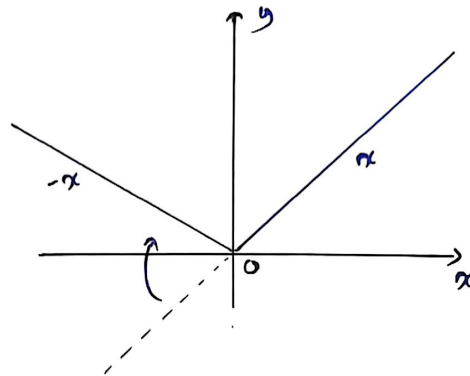
The modulus (or absolute value) of a real number represents its distance from zero on the number line.

$$y = |f(x)|$$

$$y = \begin{cases} f(x), & f(x) \geq 0 \\ -f(x), & f(x) < 0 \end{cases}$$

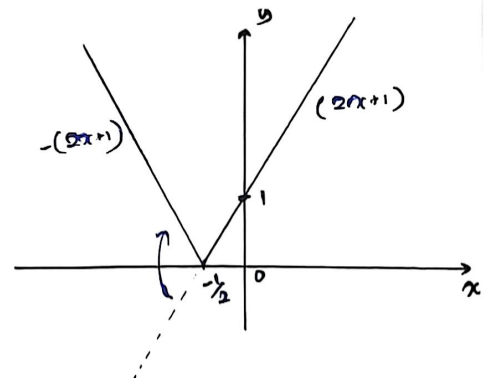
$$y = |x|$$

$$y = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$



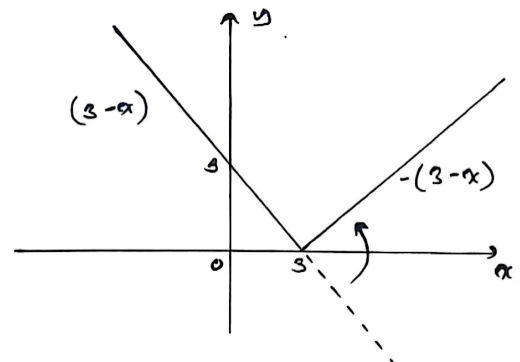
$$y = |2x + 1|$$

$$y = \begin{cases} (2x+1), & (2x+1) \geq 0, x \geq -\frac{1}{2} \\ -(2x+1), & (2x+1) < 0, x < -\frac{1}{2} \end{cases}$$



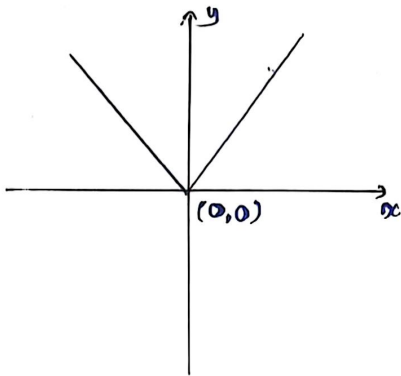
$$y = |3 - x|$$

$$y = \begin{cases} (3-x), & (3-x) \geq 0, 3 \geq x \\ -(3-x), & (3-x) < 0, 3 < x \end{cases}$$

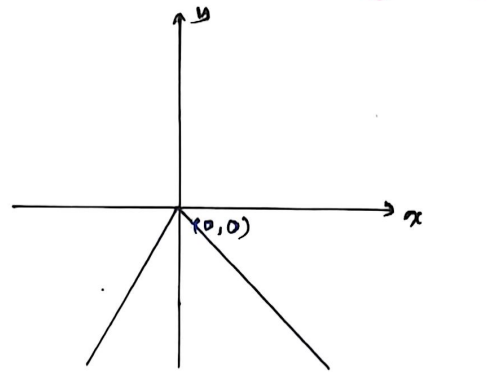


මාපාංක ශ්‍රිත පරිණාමනය කිරීම (Transforming Modulus Functions)

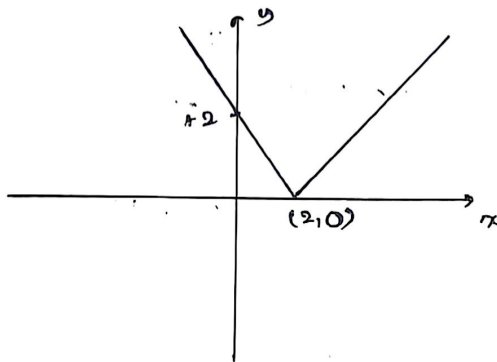
$$y = |x|$$



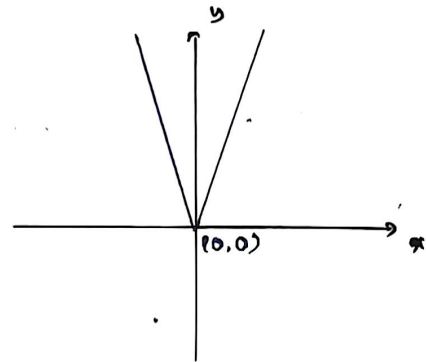
$$y = -|x|$$



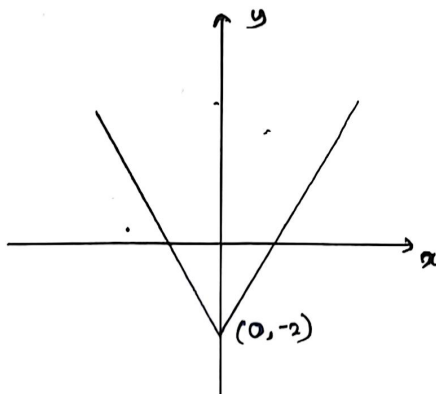
$$y = |x - 2| \quad 'x' \rightarrow (x - 2)$$



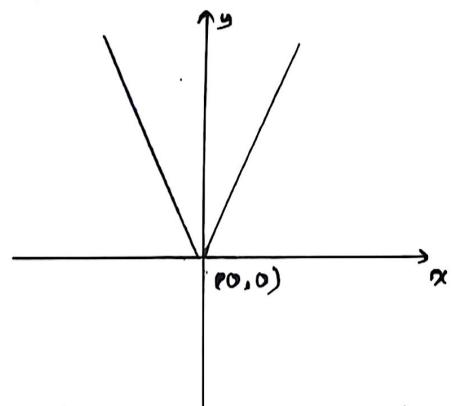
$$y = |2x| \quad 'x' \rightarrow \left(\frac{x}{2}\right)$$



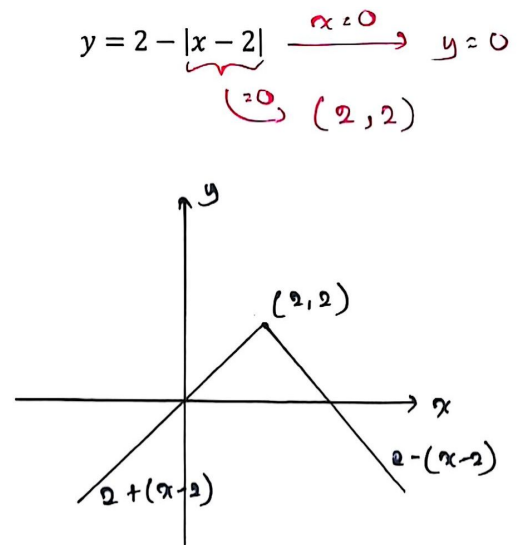
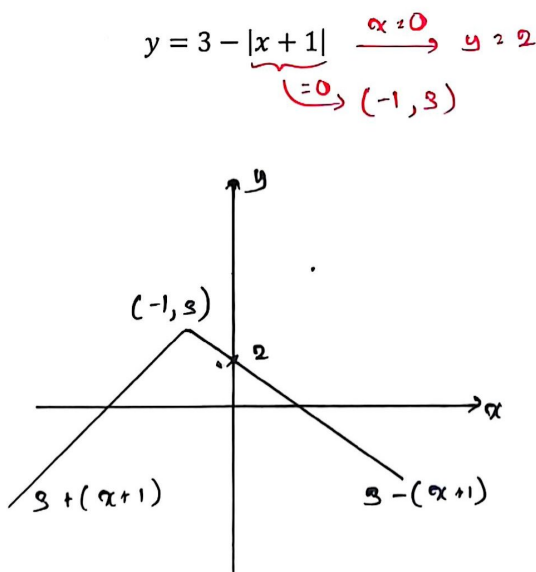
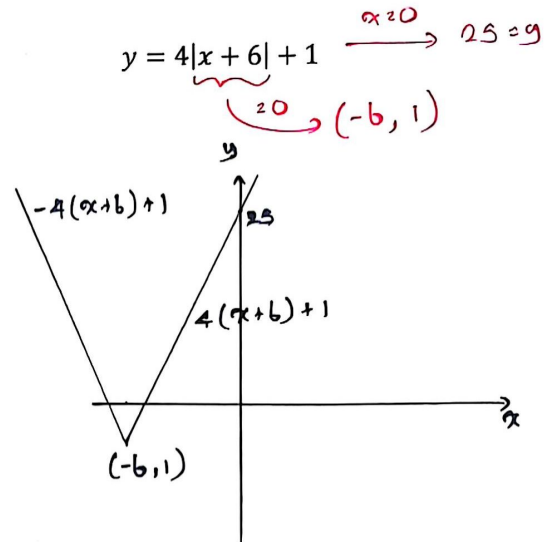
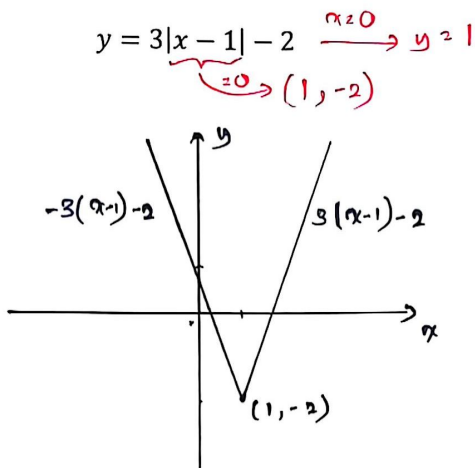
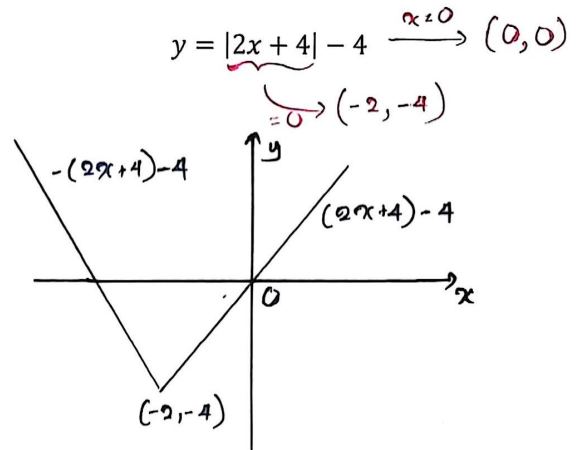
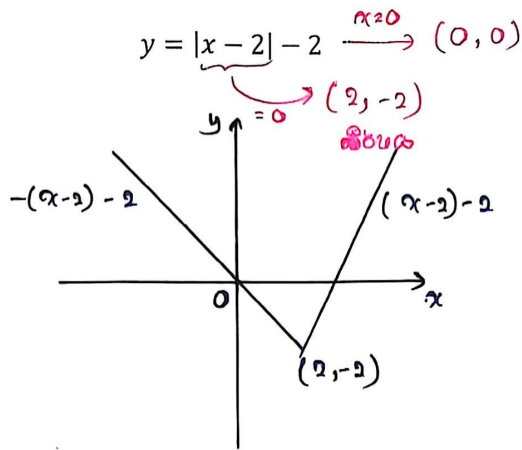
$$y = |x| - 2 \quad 'y' \rightarrow (y - 2)$$



$$y = 2|x| \quad 'y' \rightarrow (y \cdot 2)$$

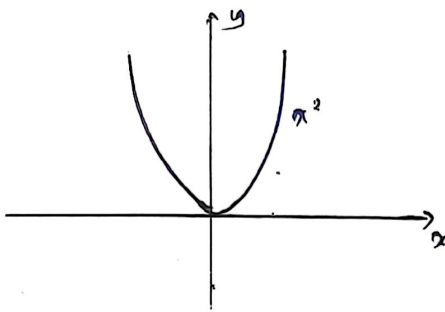


මාපාංක ශ්‍රිතවල ප්‍රස්ථාර ඇඳීම (Sketching Modulus Functions)

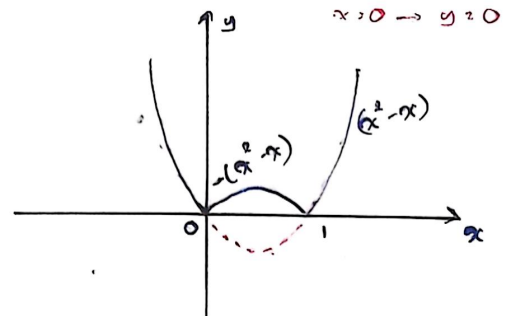


විවිධ ශ්‍රිතවල මාපාංකය (Modulus of Different Functions)

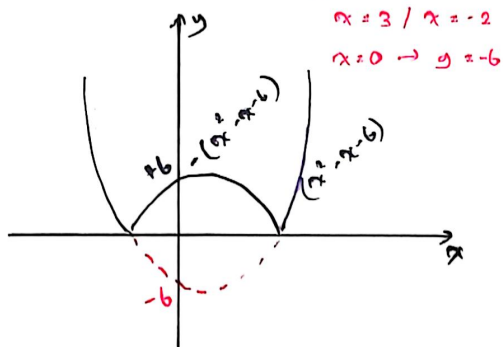
$$y = |x^2|$$



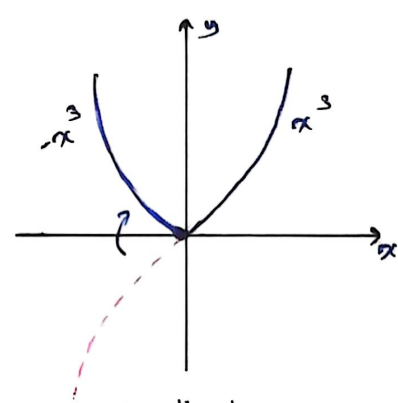
$$y = |x^2 - x|$$



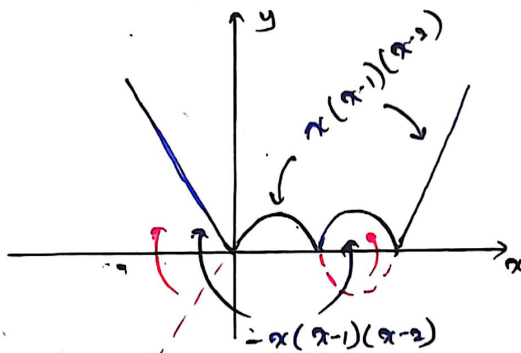
$$y = |x^2 - x - 6|$$



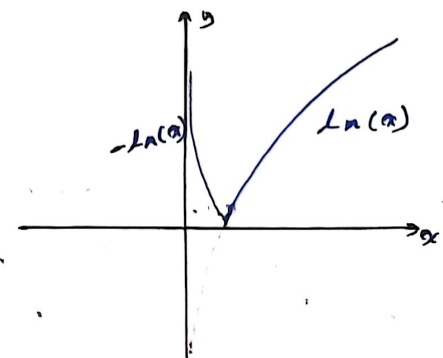
$$y = |x^3|$$



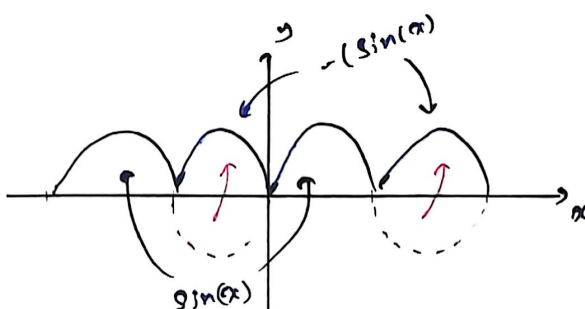
$$y = |x(x-1)(x-2)|$$



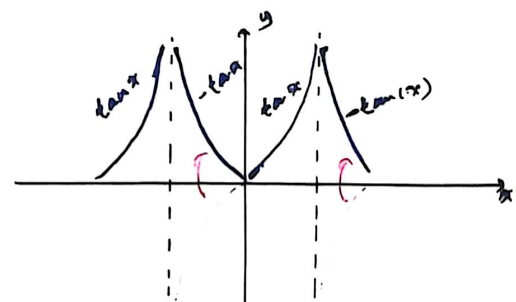
$$y = |\ln x|$$



$$y = |\sin x|$$



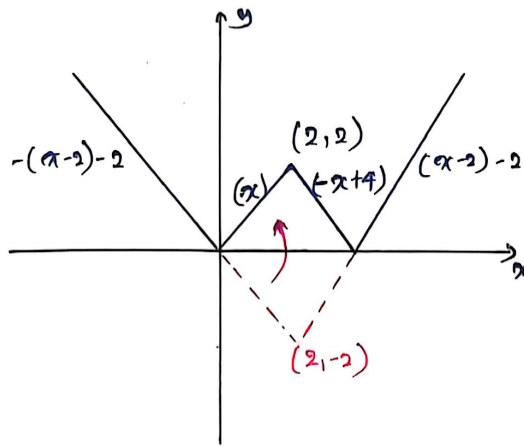
$$y = |\tan x|$$



මාපාංක ශ්‍රිතයක මාපාංකය (Modulus of a Modulus Function)

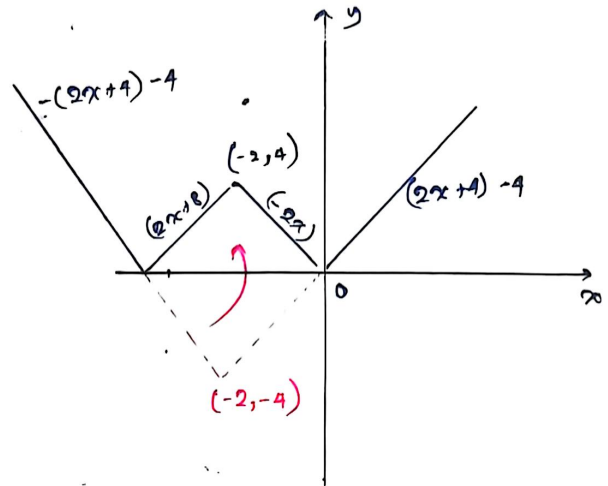
$$y = ||x - 2| - 2| \xrightarrow{x \geq 0} y \geq 0$$

$\downarrow x=0$
 $(2, -2)$



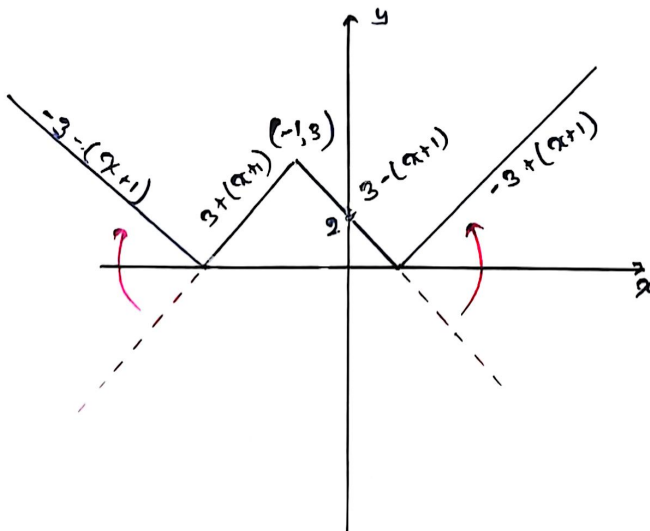
$$y = ||2x + 4| - 4| \xrightarrow{x \geq 0} y \geq 0$$

$\downarrow x=0$
 $(-2, -4)$



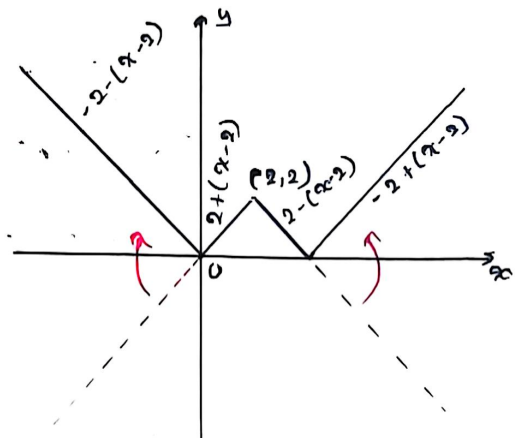
$$y = |3 - |x + 1|| \xrightarrow{x \geq 0} 2 = y$$

$\downarrow x=0$
 $(-1, 3)$



$$y = |2 - |x - 2|| \xrightarrow{x \geq 0} y \geq 0$$

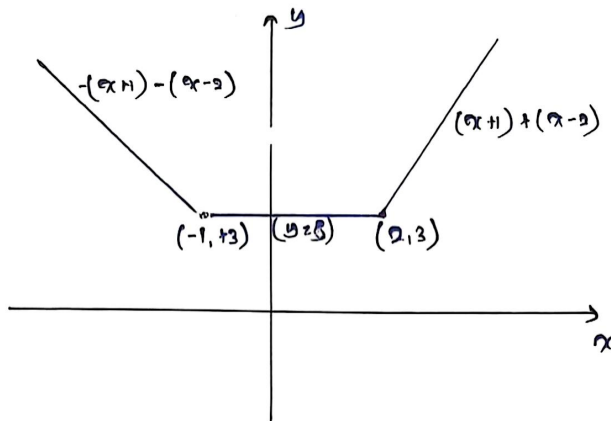
$\downarrow x=0$
 $(2, 2)$



මාපාංක ශ්‍රිත දෙකක එකතුව සහ අන්තරය (Addition & Subtraction of Two Modulus Functions)

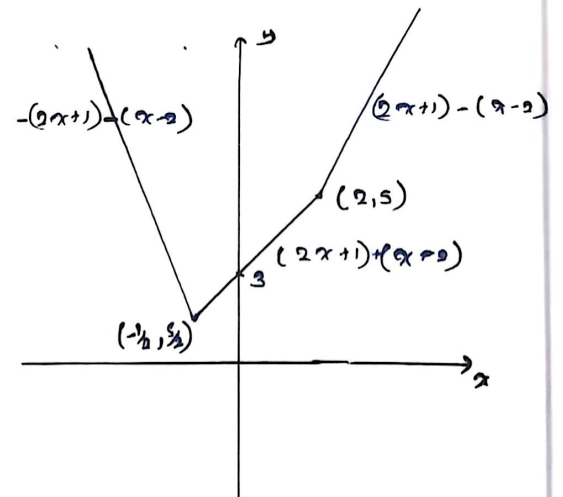
$$y = |x+1| + |x-2| \xrightarrow{x \neq 0} y = 3$$

$\downarrow x=0$ $\downarrow x=0$
 $x=-1$ $(2, 3)$
 $(-1, 43)$



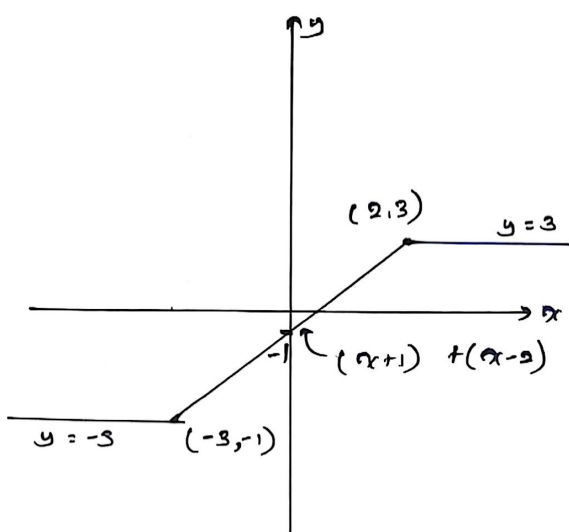
$$y = |2x+1| + |x-2| \xrightarrow{x \neq 0} y = 3$$

$\downarrow x=0$ $\downarrow x=0$
 $(-\frac{1}{2}, \frac{5}{2})$ $(2, 5)$



$$y = |x+1| - |x-2| \xrightarrow{x \neq 0} y = -1$$

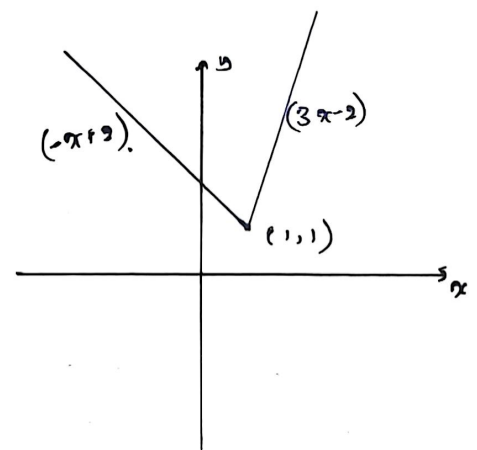
$\downarrow x=0$ $\downarrow x=0$
 $(-1, -3)$ $(2, 3)$



$$y = x + 2|x-1| \xrightarrow{x \neq 0} y = 1$$

$\downarrow x=0$
 $(1, 1)$

$x + (2x-2) = 3x-2$
 $x - (2x-2) = -x+2$



මාපාංක සමීකරණ විසඳීම (Solving Modulus Equations)

$$|x| = 3$$

$$x = \pm 3$$

$$\underline{x = 3}$$

$$\underline{x = -3}$$

$$|3x| = 9$$

$$3x = \pm 9$$

$$3x = 9$$

$$\underline{x = 3}$$

$$3x = -9$$

$$\underline{x = -3}$$

$$|x - 2| = 4$$

$$x - 2 = \pm 4$$

$$x - 2 = 4$$

$$\underline{x = 6}$$

$$x - 2 = -4$$

$$\underline{x = -2}$$

$$|3x - 1| = 5$$

$$3x - 1 = \pm 5$$

$$3x - 1 = 5$$

$$\underline{x = 2}$$

$$3x - 1 = -5$$

$$\underline{x = -\frac{4}{3}}$$

$$\left| \frac{x - 5}{2} \right| = 1$$

$$\frac{x - 5}{2} = \pm 1$$

$$x - 5 = 2$$

$$\underline{x = 7}$$

$$x - 5 = -2$$

$$\underline{x = 3}$$

$$\left| \frac{x}{6} - 1 \right| = 3$$

$$\frac{x}{6} - 1 = \pm 3$$

$$\frac{x}{6} = 4$$

$$\underline{x = 24}$$

$$\frac{x}{6} = -2$$

$$\underline{x = -12}$$

$$|3x - 4| = 9 - 2x$$

1 වන පියවර

$$3x - 4 = \pm(9 - 2x)$$

$$3x - 4 = 9 - 2x$$

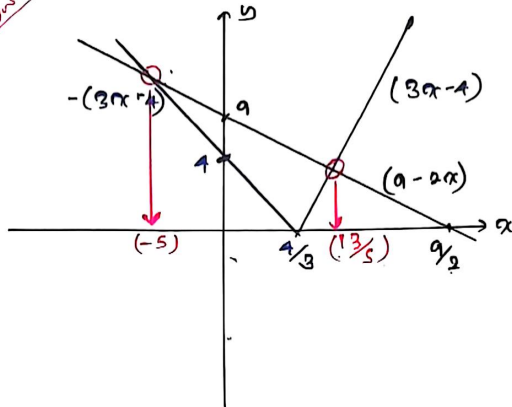
$$5x = 13$$

$$x = \frac{13}{5}$$

$$3x - 4 = -9 + 2x$$

$$x = -5$$

2 වන පියවර



$$|2x + 9| = 14 - x$$

1 වන පියවර

$$2x + 9 = \pm(14 - x)$$

$$2x + 9 = 14 - x$$

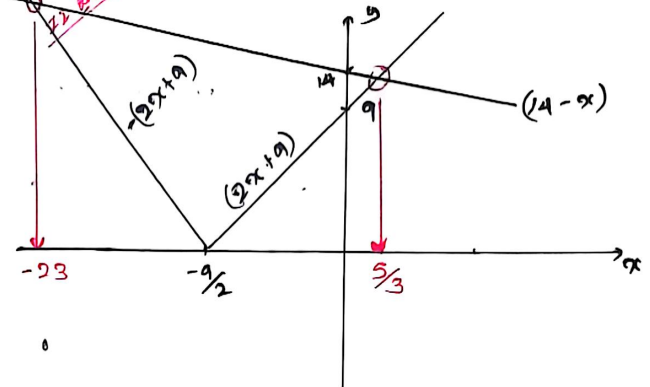
$$3x = 5$$

$$x = \frac{5}{3}$$

$$2x + 9 = -14 + x$$

$$x = -23$$

2 වන පියවර



$$|6 - x| = \frac{1}{2}x - 5$$

1 වන පියවර

$$6 - x = \pm\left(\frac{1}{2}x - 5\right)$$

$$6 - x = \frac{1}{2}x - 5$$

$$12 - 2x = x - 10$$

$$3x = 22$$

$$x = \frac{22}{3}$$

$$\frac{1}{2}x - 5 \geq 0$$

$$x \geq 10$$

$$6 - x = -\frac{1}{2}x + 5$$

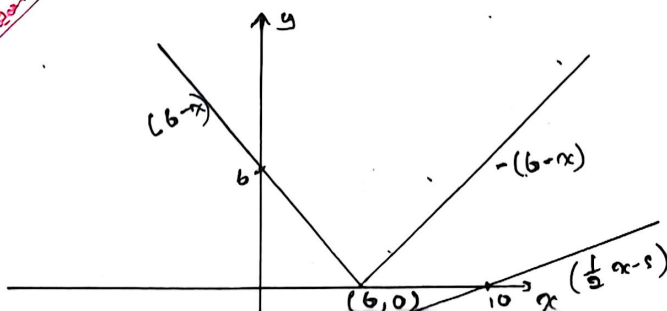
$$12 - 2x = -x + 10$$

$$-x = -2$$

$$x = 2$$

$$x \geq 10$$

2 වන පියවර



$$\left|\frac{1}{2}x - 2\right| = -2x$$

1 වන පියවර

$$\frac{1}{2}x - 2 = \pm(-2x)$$

$$\frac{1}{2}x - 2 = -2x$$

$$x - 4 = -4x$$

$$5x = 4$$

$$x = \frac{4}{5}$$

$$\frac{1}{2}x - 2 = 2x$$

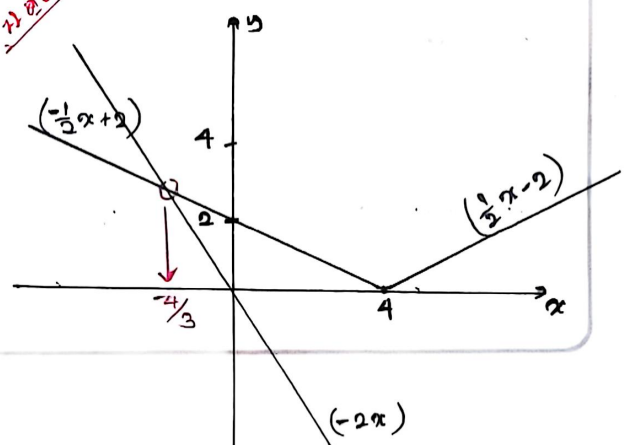
$$x - 4 = 4x$$

$$3x = -4$$

$$x = -\frac{4}{3}$$

$$-2x \geq 0 \text{ බව සහතිකය.}$$

2 වන පියවර

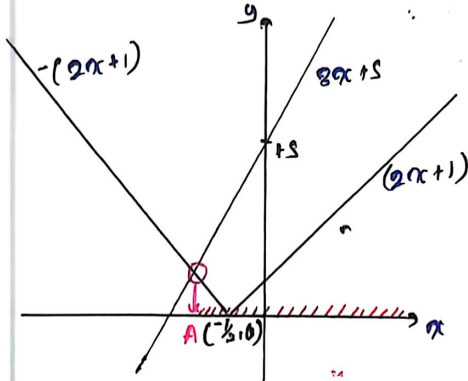


මාපාංක අසමානතා විසඳීම (Solving Modulus Inequalities)

$$(0, 1) \xleftarrow{x \geq 0} |2x+1| < 3x+5 \xrightarrow{x \geq 0} (0, 5)$$

$$\downarrow x=0 \quad \downarrow x=0$$

$$(-\frac{1}{2}, 0) \quad (-\frac{5}{3}, 0)$$



$$A: -2x-1 < 3x+5$$

$$5x > -6$$

$$x > (-\frac{6}{5})$$

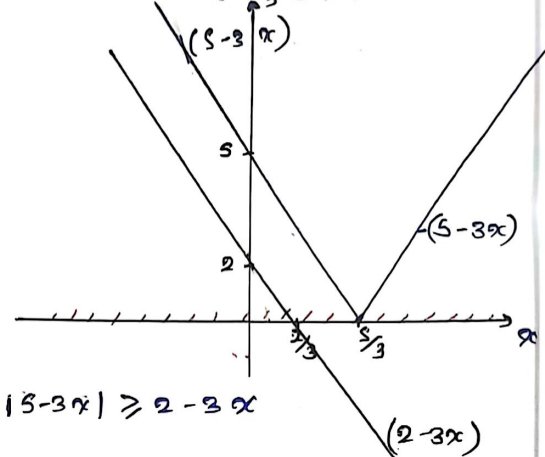
$$|2x+1| < 3x+5$$

$$\underline{x > -\frac{6}{5}}$$

$$(0, 5) \xleftarrow{x \geq 0} |5-3x| \geq 2-3x \xrightarrow{x \geq 0} (0, 2)$$

$$\downarrow x=0 \quad \downarrow x=0$$

$$(\frac{5}{3}, 0) \quad (\frac{2}{3}, 0)$$



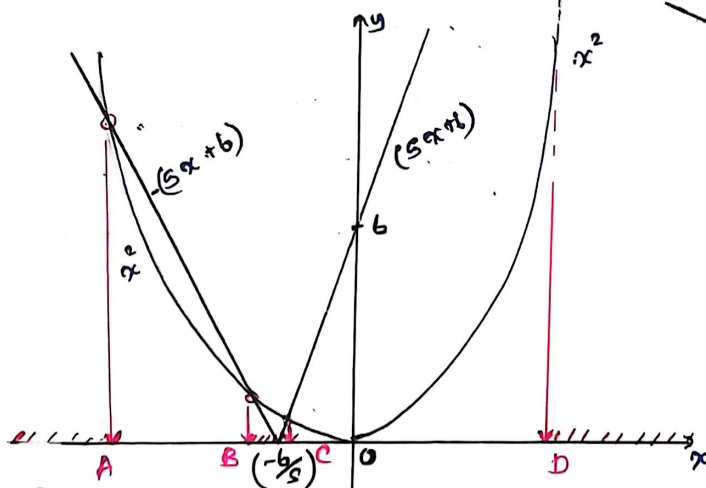
$$|5-3x| \geq 2-3x$$

$$\underline{x \in \mathbb{R}}$$

$$x^2 > |5x+6| \xrightarrow{x \geq 0} (0, 6)$$

$$\downarrow x=0$$

$$(-\frac{6}{5}, 0)$$



$$B, A: x^2 < -5x-6$$

$$x^2+5x+6 > 0$$

$$(x+3)(x+2) > 0$$

$$x < -3 / x < -2$$

$$C, D: x^2 < 5x+6$$

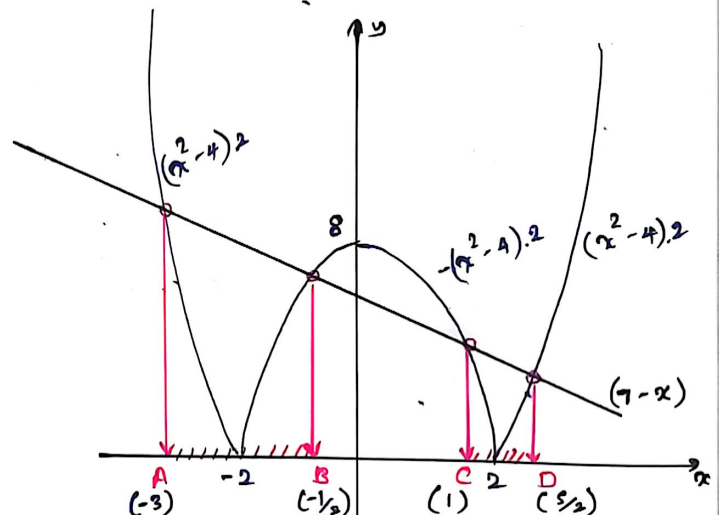
$$x^2-5x-6 < 0$$

$$(x+1)(x-6) < 0$$

$$x > -1 / x < 6$$

$$\underline{x \in (-\infty, -3) \cup (-2, -1) \cup (6, \infty)}$$

$$7-x \geq 2|x^2-4|$$



$$A, D: 2(x^2-4)^2 \geq 7-x$$

$$2x^2+x-15 \geq 0$$

$$(x+3)(x-\frac{5}{2}) \geq 0$$

$$x < -3 / x \geq \frac{5}{2}$$

$$B, C: -2(x^2-4)^2 \geq 7-x$$

$$-2x^2+8 = 7-x$$

$$2x^2-x-1 < 0$$

$$(x+\frac{1}{2})(x-1) < 0$$

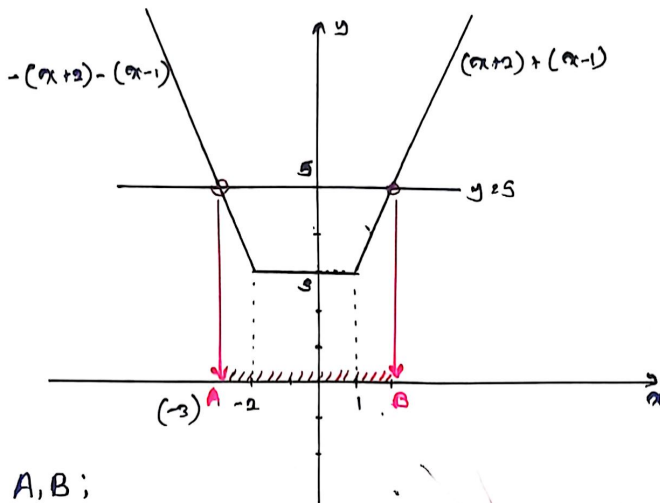
$$x > -\frac{1}{2} / x < 1$$

$$\underline{x \in [-3, -\frac{1}{2}] \cup [1, \frac{5}{2}]}$$

$$(0, 3) \leftarrow x \geq 0 \quad |x+2| + |x-1| < 5$$

$$\downarrow x=0 \quad \downarrow x=0$$

$$(-2, 3) \quad (1, 3)$$



A, B;

$$-(x+2) - (x-1) \geq 5$$

$$-2x - 1 \geq 5$$

$$x \leq -3$$

$$(x+2) + (x-1) \geq 5$$

$$2x + 1 \geq 5$$

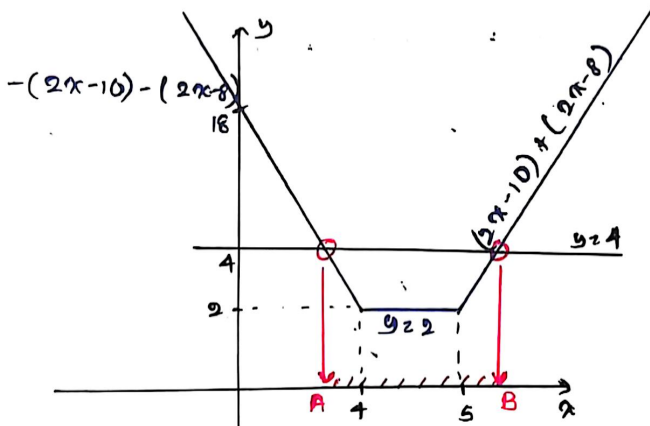
$$x \geq 2$$

$$\underline{-3 < x < 2}$$

$$(0, 18) \leftarrow x \geq 0 \quad |2x-10| + |2x-8| \leq 4$$

$$\downarrow x=0 \quad \downarrow x=0$$

$$(5, 2) \quad (4, 2)$$



$$A; -(2x-10) - (2x-8) = 4$$

$$-4x + 18 = 4$$

$$x = \frac{7}{2}$$

$$\underline{\frac{7}{2} \leq x \leq \frac{11}{2}}$$

$$B; (2x-10) + (2x-8) = 4$$

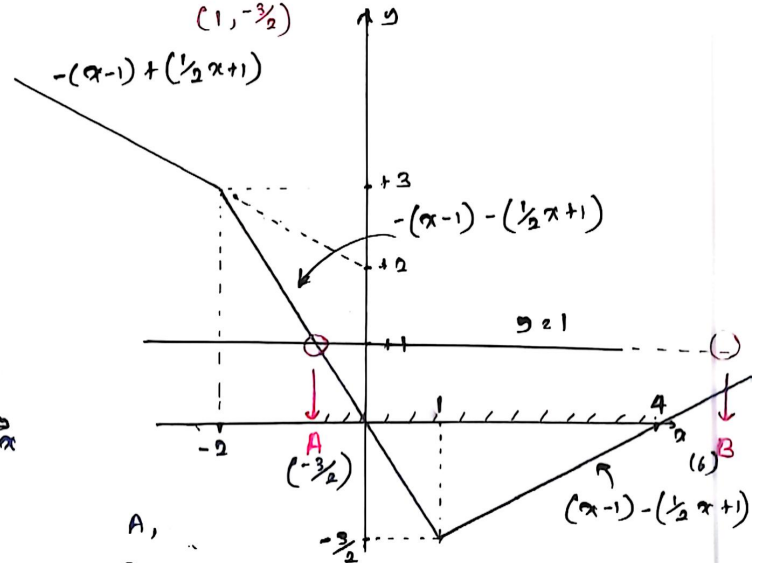
$$4x - 18 = 4$$

$$x = \frac{11}{2}$$

$$(0, 0) \leftarrow x \geq 0 \quad |x-1| - \left| \frac{1}{2}x + 1 \right| < 1$$

$$\downarrow x=0 \quad \downarrow x=0$$

$$(1, -\frac{3}{2})$$



A,

$$-\frac{3}{2}x = 1$$

$$x = -\frac{2}{3}$$

B,

$$\frac{1}{2}x - 2 = 1$$

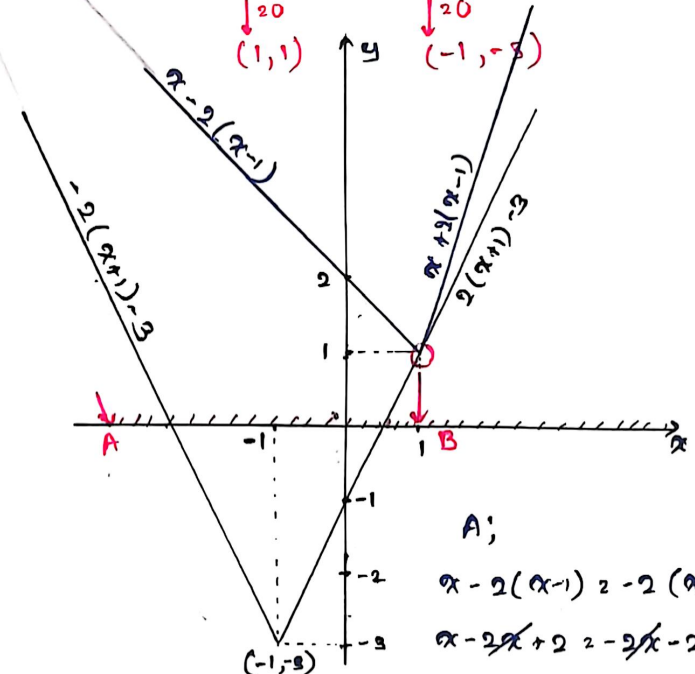
$$x = 6$$

$$\underline{-\frac{2}{3} < x < 6}$$

$$(0, 2) \leftarrow x \geq 0 \quad x + 2|x-1| > 2|x+1| - 3 \xrightarrow{x \geq 0} (-1) = 9$$

$$\downarrow x=0 \quad \downarrow x=0$$

$$(1, 1) \quad (-1, -3)$$



A;

$$x - 2(x-1) = 2(x+1) - 3$$

$$x - 2x + 2 = 2x + 2 - 3$$

$$x = -7$$

$$\underline{x \in (-7, 1) \cup (1, \infty)}$$

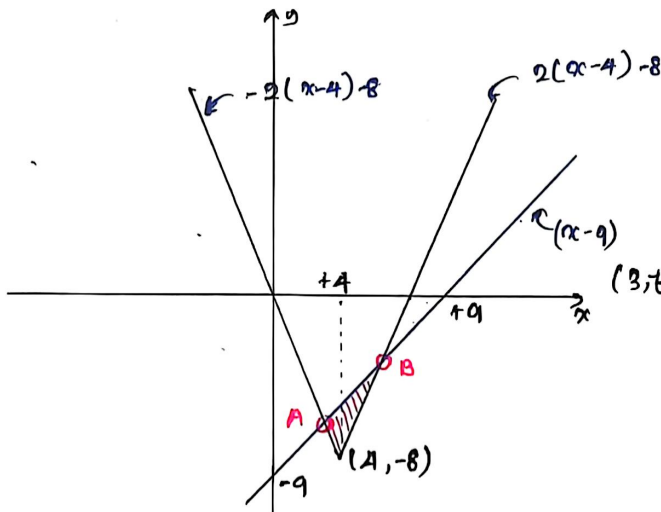
මාපාංක ශ්‍රිත ආවෘත කරන ප්‍රදේශවල වර්ගඵල සෙවීම
(Finding Area of the Regions enclosed by the Modulus Functions)

$y = 2|x - 4| - 8$ සහ $y = x - 9$ ට අදාළ ප්‍රස්ථාර ආවෘත කරන ප්‍රදේශයේ වර්ගඵලය වර්ග ඒකක 6 ක් බව පෙන්වන්න.

Show that the area of the shaded region that is enclosed by $y = 2|x - 4| - 8$ and $y = x - 9$ is 6.

$(0,0) \xleftarrow{x=0} y = 2|x-4|-8$ / $y = x-9 \xrightarrow{x=0} (0,-9)$
 $\downarrow = 0$ $\downarrow = 0$
 $(4, -8)$ $(9, 0)$

$\begin{aligned} A; -2(x-4)-8 &= x-9 \\ -2x+8-8 &= x-9 \\ -2x &= x-9 \\ x &= 3 \\ y &= -6 \end{aligned}$	$\begin{aligned} B; 2(x-4)-8 &= x-9 \\ 2x-8-8 &= x-9 \\ 2x-16 &= x-9 \\ x &= 7 \\ y &= -2 \end{aligned}$
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Shoelace method

$$A = \frac{1}{2} \begin{vmatrix} 3 & 4 & 7 \\ -6 & -8 & -2 \end{vmatrix}$$

$$A = \frac{1}{2} [(-24 - 56 - 6) - (-24 - 8 - 42)]$$

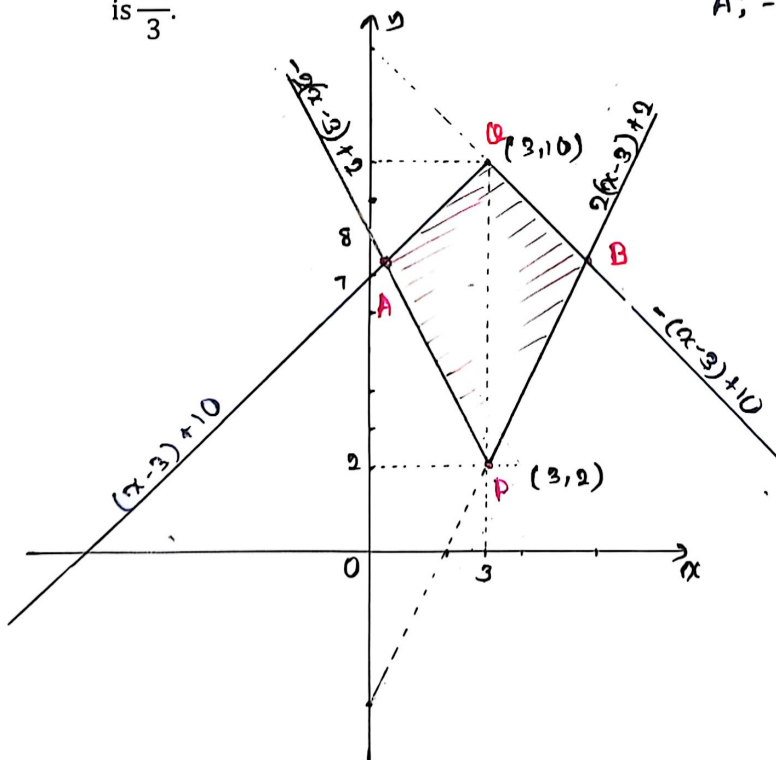
$$A = \frac{1}{2} [-12]$$

$$A = \frac{1}{2} \times 12 \rightarrow \underline{\underline{A = 6}}$$

$y = -|x - 3| + 10$ සහ $y = 2|x - 3| + 2$ ට අදාළ ප්‍රස්ථාර ආවෘත කරන ප්‍රදේශයේ වර්ගඵලය වර්ග ඒකක $\frac{64}{3}$ ක් බව පෙන්වන්න.

Show that the area of the shaded region that is enclosed by $y = -|x - 3| + 10$ and $y = 2|x - 3| + 2$

is $\frac{64}{3}$.



$\begin{aligned} A; -2(x-3)+2 &= (x-3)+10 \\ -2x+6 &= x-3+10 \\ -2x &= x+7 \\ x &= -\frac{7}{3} \\ y &= \frac{22}{3} \end{aligned}$	$\begin{aligned} B; 2x-4 &= -x+13 \\ 3x &= 17 \\ x &= \frac{17}{3} \\ y &= \frac{22}{3} \end{aligned}$
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• shoelace method

$$\text{Area} = \frac{1}{2} \begin{vmatrix} -\frac{7}{3} & \frac{17}{3} & 3 \\ \frac{22}{3} & \frac{22}{3} & 10 \end{vmatrix}$$

$$\text{Area} = \frac{1}{2} \left[\left(22 + \frac{34}{3} + 22 + \frac{10}{3} \right) - \left(\frac{22}{3} + 22 + \frac{170}{3} + 22 \right) \right]$$

$$\text{Area} = \frac{1}{2} \left[\left(\frac{176}{3} \right) - \left(\frac{304}{3} \right) \right]$$

$$\text{Area} = \frac{1}{2} \left[-\frac{128}{3} \right]$$

$$\text{Area} = \frac{128}{3} \times \frac{1}{2} = \underline{\underline{\frac{64}{3}}}$$